

1. The symbol to the right represents:

- A) Radiation
- B) Biohazard
- C) Chemical hazard
- D) Heat hazard



2. Which one of the following is not used for its morphological aspect of diagnosis

- A) Serology
- B) Gram stain
- C) Giemsa stain
- D) AFB stain
- E) Wright stain

3. Which one of the following microbes is not stained via gram stain?

- A) Staphylococci
- B) Streptococcus
- C) Spirochete
- D) Coccobacilli
- E) Vibrio

1. The trophozoite, schizont and gametocyte stages are usually present in the peripheral blood of all plasmodium species except:

- A) Plasmodium falciparum
- B) Plasmodium malarae
- C) Plasmodium vivax
- D) Plasmodium ovalae

2. Under thin smear preparation you observe stacked up RBC. What is the most probable cause:

- A) Due to time delay of blood sample before smearing
- B) Because of usage of outdated anticoagulant
- C) Due to contamination of the smear by insects

3. All of the following equipment's are used for thin smear preparation except:

- A) Microscopic slide
- B) Physiological saline
- C) Applicator stick
- D) Wax pencil
- E) Microscope

4. Which one of the following is not true a true feature about Weil – Felix test?

- A) Antigen Ox19 – p. vulgaris
- B) Antigen Ox2 – p. vulgaris
- C) Antigen O_{xk} – p. mirabilis
- D) We apply it for the diagnosis of typhoid

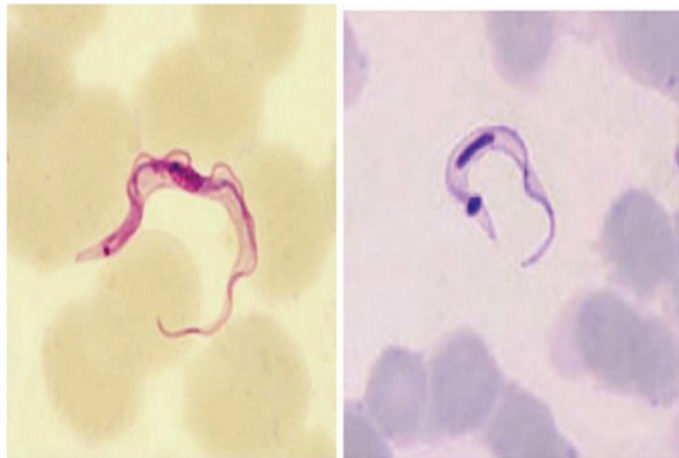
5. Which one of the following is correct regarding germ tube test
- A) Used to differentiate candida albicans from non- candida species
 - B) Discovered by Christian gram
6. Which one of the following methods is used for a rapid pathological diagnosis of patients at health care
- A) 30% Giemsa stain solution applied for 40 – 60 minutes
 - B) 3% Giemsa stain solution applied for 40 – 60 minutes
 - C) 10% wright stain solution applied for 15 minutes
 - D) 10% Giemsa stain solution applied for 15 minutes
7. False about the India ink stain
- A) The stain is negative
 - B) The stain is positive
 - C) The halo seen on the picture is it's capsule
 - D) The species seen causes cryptococcus meningitis
8. The most common vein from which blood is taken from in venipuncture is:
- A) Median cubital vein
 - B) Basilic vein
 - C) Cephalic vein
 - D) A and B
9. Among blood sample taking techniques in which method is tissue fluid contamination a risk?
- A) Capillary
 - B) Venipuncture
 - C) Artery
 - D) A and B
 - E) A and C
10. During the venipuncture blood sample taking procedure the purpose of the needle being beveled up serves us what function
- A) To prevent hematoma formation
 - B) To control volume of blood flow
 - C) To prevent from mixing up with tissue fluid
11. The purpose of using KOH preparation is:
- A) To remove keratinized material from the specimen
 - B) To kill and preserve the bacteria
 - C) To decolorize the stain applied to the specimen
12. Which one of the following is correct regarding gram staining:
- A) Crystal violet – mordant
 - B) Iodine – first dye
 - C) Neutral red – decolorizing agent
 - D) Safranin - counter stain
 - E) Alcohol – stains the cell wall

13. During gram staining both bacteria and fungi are stained positive. What is the characteristic feature to differentiate between the two:
- A) size
 - B) color
 - C) pink color
 - D) blue color
14. Which one of the following is correct regarding Giemsa stain?
- A) Giemsa stain is a differential stain since it can stain the different parts of the parasite separately
 - B) Giemsa stain is a differential stain since some bacteria are Giemsa positive while others are Giemsa negative
15. Which of the following plasmodium species form a characteristic band over the infected RBC
- A) Plasmodium falciparum
 - B) Plasmodium malarae
 - C) Plasmodium vivax
 - D) Plasmodium ovalae
16. Which of the following plasmodium species have shuffner spots on their respective infected RBCs
- A) Plasmodium vivax
 - B) Plasmodium ovalae
 - C) Plasmodium knowlesi
 - D) A and C
 - E) A and B
17. From the image below identify the species (A), (B) and (C) – disease

(A) _____

(B) _____

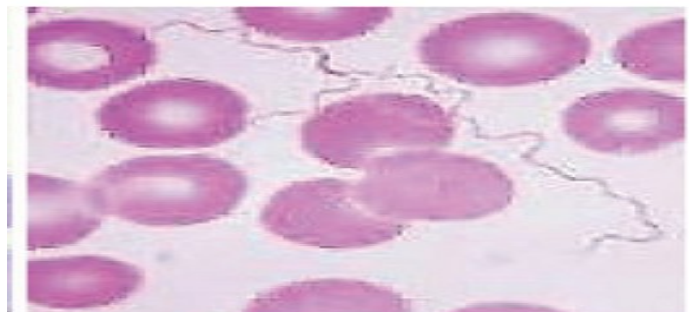
(C) _____



18. From the image below identify the species name and the disease it causes:

The species name: _____

The disease caused by the species: _____



19. The most common characteristics of bacteria include

- A) mesophilic
- B) psychrophilic
- C) thermophilic
- D) neutrophilic
- E) A and D

20. false about s. pyogenes

- A) optochin sensitive
- B) bacitracin sensitive
- C) arranged in chains

21. false about s. pneumoniae

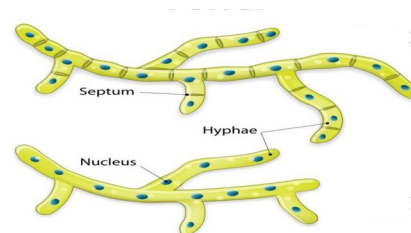
- A) bacitracin sensitive
- B) optochin sensitive
- C) causes cap
- D) causes alpha hemolysis

22. which one of the following culture media is not used for growth of fungi species

- A) Blood agar
- B) Mackonkey agar
- C) SDA
- D) Mycel agar
- E) Dermatophyte agar

23. which one of the following is false

- A) the organism above is a septate hyphae
- B) the organism below is aseptate hyphae
- C) the organism is hyphal stage of fungi
- D) the organism is a bacteria

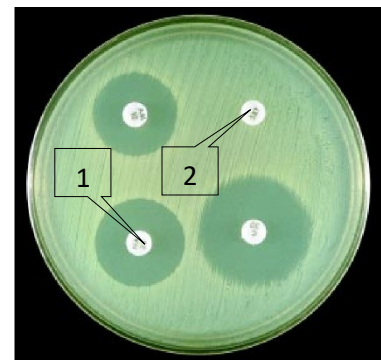


24. which one of the following is incorrect regarding the lactophenol cotton blue stain:

- A) phenol is used to kill the fungi
- B) lactic acid is used to fix and keep the morphology of the fungi intact
- C) cotton blue is used to stain the chitin cell wall of the fungi
- D) cotton blue is used to stain the peptidoglycan cell wall of the fungi

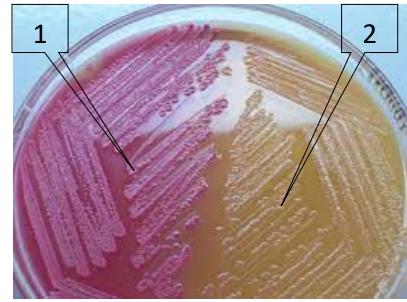
25. which one of the following is incorrect based on the disc diffusion method presented on the image below:

- A) 1 is sensitive
- B) 2 is resistant to the disc
- C) it's minimally inhibitory method
- D) it's a disk diffusion method



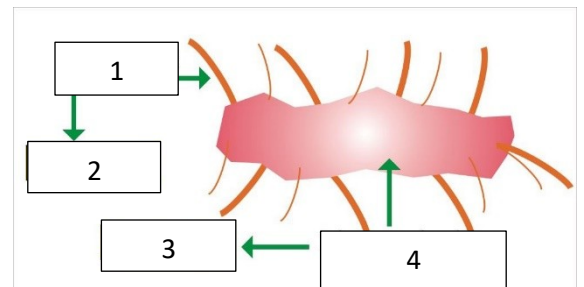
26. which one of the following statements is incorrect with regard to the picture below:

- A) 1 is lactose fermenter
- B) 2 is non lactose fermenter
- C) *E. coli* is a lactose fermenter
- D) *klebsiella* is a lactose fermenter
- E) the agar contains lactose as an ingredient



27. All of the following describes the characteristics of Widal test except:

- A) 1 represents the flagella of the salmonella typhi
- B) 4 represents the cell wall of salmonella typhi
- C) 2 represent the flagellar H antigen
- D) 3 represents the cell wall antigen O_x -2
- E) We use this test for the diagnosis of typhoid



28. which one of the following laboratory equipment is a volumetric ware?

(A)



(C)



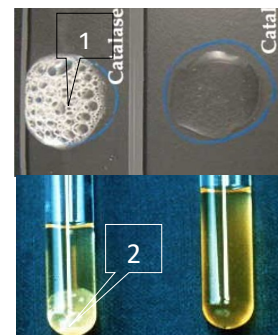
(B)



(D)



29. Under thick smear preparation slide you were able to observe clear and numerous intact RBC's. what is the probable cause behind this phenomenon?
- Because of application of methanol usage during the smear preparation
 - Due to coagulation of the blood sample used for preparation
 - The parasite that is present in the smear can clump them together and make them visible
 - Because of heat fixation of the thick smear
30. which one of the following cell types are not observe under thick smear preparation slide?
- RBC
 - white blood cells
 - platelets
 - granules of the parasites
31. consider you are taking a urine sample from a UTI patient to diagnose via urine dipstick test, which one of the following diagnostic criteria is not measured by this test?
- nitrite
 - protein
 - bilirubin
 - lymphocyte esterase
32. which of the following factors has no great impact in the process of proper diagnosis of microbes via laboratory methods
- the specimen taken is ideal for the diagnosis
 - the volume of specimen taken
 - the technique used to process the specimen sample
 - the length of the diagnostic method
33. Which one of the following species is not stained via AFS
- Nocardia
 - Cryptosporidium
 - Staphylococci
 - M. leprae
 - M. tuberculosis
34. Which one of the following is incorrect based on the ff image:
- 1 is catalase positive
 - 1 is coagulase is positive
 - Coagulase -ve - S. aureus
 - Coagulase -ve - CONS



35. During the thin smear preparation, the following thin smear was obtained. What is the probable cause of this phenomenon



- A) Presence of grease in the slide
- B) The edges of the spreading microscopic slide is broken
- C) The speed of the spreading process was not well controlled
- D) The smear didn't cover the adequate amount of space

36. Which one of the following is not included under Amsel's criteria:

- A) Characteristic vaginal discharge
- B) Vaginal PH >4.5
- C) Positive whiff test
- D) Presence of clue cells
- E) Is used to diagnose morphotypes of causative agents

37. Urinary is not the clinical finding of urine test

- A) <10,000/ml of urine is considered in significant
- B) 10,000 – 100,000/ml of urine is considered clinically significant infection
- C) >100,000/ml of urine implicates clinically severe infection
- D) The midstream urine is the ideal specimen for sampling
- E) CLED agar is a differential medium that is only suitable for the growth of fungi

38. From the figure

Acid fast red pink

Blue non afb

39. Which of the following plasmodium species has amoeboid trophozoite stage?

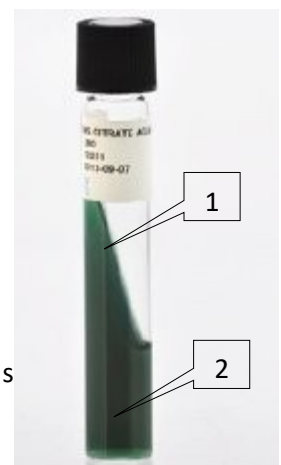
- A) Plasmodium malarae
- B) Plasmodium vivax
- C) Plasmodium falciparum
- D) Plasmodium ovalae

40. which one of the following statements is incorrect Based on the image below?

- A) 1 represents the slant
- B) 2 represents the bottom
- C) It's positive
- D) The test tube contains citrate agar
- E) Its source is carbon (energy)

41. Which one of the following features are not reported when examining a gram s

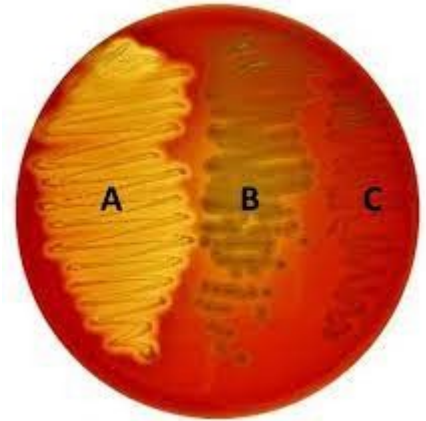
- A) Number of parasites



- B) Number of pus cells in the smear
 - C) Morphology of the parasites
 - D) The etiologic agent of the disease
42. Which one of the following is incorrect regarding gram stain:
- A) Simple stain
 - B) Differential stain
 - C) Multistep process
 - D) Under the gram stain bacteria are stained differently as gram positive and gram negative
43. Which one of the following safety measures are taken during a laboratory procedure?
- A) Let down long hair to dangle over the shoulders
 - B) Splashing and spilling in the laboratory
 - C) Put your proper safety glasses on

44. Choose the incorrect statement according to the picture below:

- A) The inoculum on A is beta hemolytic species
- B) The inoculum on B is alpha hemolytic species
- C) The culture medium is muller Hinton agar
- D) The species on A could be streptococcus pyogenes



45. Which of the following is incorrectly matched?

- A) Enriched medium – nutrient agar
- B) Enriched media – blood agar and chocolate agar
- C) Enrichment agar – blood agar
- D) Differential medium – MacConkey agar